Binary transfer classification and obstruction audit

Date: 2026-10-30. Status: implemented bounded audit and written scoped theorems; geometric formalization and universal efficient-transfer classification remain partial.

Question and actual deliverable

The user requested a classification/trust theorem and obstruction, with a Rust CLI whose coverage certificates make clear whether an input curve and its isogeny class have been exhausted. The result is a replayable tool with three different scopes: abstract finite-group maps, algebraic homomorphisms of abelian varieties, and evaluated maps in a specified implementation family. They must not be conflated. None is an exhaustive classification of all useful ECDLP algorithms. No new weak family or solver-speed claim is made.

The implementation is based on main ae8994ab14e942c774818704dc19402f6671d848, on branch codex/transfer-audit-classification; main 82eb1eac was subsequently integrated without changing the imported native sources or tool implementation. Native cover/arithmetic sources are imported unchanged by path from suite/src/binary_ecc. The old GHS research files, source snapshots, PDFs and manifests are untouched.

Results and theorem scope

For an abstract cyclic group of prime order r and finite abelian group H, Hom(C_r,H) is parameterized by H[r], and a nonzero map is injective. Existence is equivalent to r dividing |H|. This says nothing about efficient evaluation: the abstract formula may require the discrete logarithm one hopes to solve.

For actual abelian varieties over a finite field, a nonzero algebraic homomorphism exists up to isogeny exactly when their Frobenius polynomials share an irreducible factor. Coprimeness excludes every algebraic homomorphism between the specified classes, not only maps generated by GHS. The CLI tests this with exact polynomial gcds, conditional on the supplied target polynomial being its actual Frobenius polynomial. It does not construct an arbitrary Jacobian, verify that polynomial geometrically, or prove nonzero action on the selected r-subgroup. The written proof also states the dimension obstruction under a separately established irreducibility hypothesis; the tool does not silently assert simplicity.

For all Jacobians over F_q of genus at most G, the exact Weil bound gives the family-independent obstruction r>floor((sqrt(q)+1)^(2G)). This applies uniformly to every source curve connected by subgroup-preserving isogenies. It needs no restriction on hyperellipticity, cover degree or automorphisms.

The full statements, written proofs, prior sources and proof map are in THEOREMS.md. They are applications of known theory, not claims of a new published classification theorem.

Vector figures follow the theorem text; editable SVG and DOT are retained.

B-163 fixed-fixture correctness checks

The public SEC 2 section 3.2.3 parameters are frozen in fixtures/b163.json. The field is defined by z^163+z^7+z^6+z^3+1; a=1, and b, G and r are retained as exact strings. This is B-163/sect163r2, not K-163/sect163k1. The fixture is an audit input, not a new EC1/IC1 curve or weak-curve catalogue entry.

Native tests verify the field, recursive Pocklington prime certificate, nonzero generator and rG=O. The source order is recovered as the unique multiple of r in the Hasse interval. The two divisor fields are enumerated: F2 and the original F_(2^163). The complete coefficient Frobenius orbit has 163 members, with evaluated point images. This is not the full isogeny class.

Over F2 the exact genus-63 upper bound is 169721301146768189648370410887560301301009728819 7, below the certified prime 584600654932361167281474244287639068925684320158 7. At genus 64 the bound is 989208235251040375755017297514670212228379369963 53. Thus genus<=63 is excluded for every target in this scope. Passing the genus-64 size test supplies no target or attack. The earlier outline's

m~30 at degree 163 is corrected: classical GHS m=163 here; 30 is the minimum for degree 233.

Vector figures follow the theorem text; editable SVG and DOT are retained.

The existing same-field genus-three family is constructed through the native backend. Tests check the reduced Mumford generator image, exact order, the degree-three pull/push identity, and the public known-answer relation at 37G. This is a checked transfer over F_(2^163), not a descent to F2. No unknown logarithm was solved, and no construction/transfer/solve timings are reported. The native module still lacks a general arbitrary-divisor norm. No new cryptanalytic experiment or research sweep was launched; these are deterministic implementation/proof/fixed-fixture correctness checks.

Coverage, validation and failures

Small-field exhaustive classification is implemented only through degree eight. It enumerates both canonical twists for every nonzero b and filters by exact order. Equal trace gives isogeny-class membership; paths between all members are not invented. Independent x,y counts through degree four control the trace counter. Tests cover compatible/incompatible spectral polynomials, malformed fields and inputs, altered prime proofs, deleted coverage rows, forged certificates and overwrite attempts. Koblitz b=1 remains a degenerate classical GHS control, while the existing same-field cover can still be valid there.

The certificate replay compares every field with a fresh deterministic calculation. It binds source, manifest, lockfile and imported native-module hashes. This is source-bound replay, not an independent implementation or authentication. `all_possible_transfers_ruled_out` remains false and open coverage is retained even after a successful run.

Lean 4.34.1 verifies eleven named theorems plus one generated equation theorem; the transitive audit accepts only propext, Classical.choice and Quot.sound. No sorry, custom claim axiom, native_decide or bypass is used in the final proof. The geometry, finite-group classification, Tate theorem and complete Rust refinement remain written/unformalized bridges. This module is tool verification under tools/lean, checked by its own CI gate; no historical corpus receipt is regenerated.

All attempts are preserved under evidence, including the initial offline dependency failure, Lean syntax/reduction failures and strict-Clippy findings. The native-module lint policy is inherited narrowly from suite/Cargo.toml; no new-code lint is disabled. The existing slow native curve-order test stays ignored as it was upstream; the new B-163 test explicitly checks its own generator's exact order. The final check receipt reports the actual checks.

The checkpoint gate first failed at incompatible BSD/GNU stat output. A one-line portable file-size query fixes it; the subsequent upload fails for missing runtime credentials/token. Those outputs are preserved. This blocks research sweeps, not the implementation and fixed-fixture correctness tests. The local LMFDB NAS/cache paths were absent; no remote catalogue was substituted. The lead atlas and existing GHS studies supply prior-work context, not novelty.

Requirement status and next discriminating work

Requirement	Status	Remaining gap
Rust CLI and replayable scope certificate	implemented and locally checked	hosted exact-head CI remains a separate gate
All subfields and exact genus obstruction	checked for the fixed fixtures	larger supported inputs require their own certificate replay
Source isogeny-class applicability	complete only for requested degree<=8 enumeration; Frobenius orbit evaluated	arbitrary large-class paths and maps remain open
Classification/trust/obstruction theorem	scoped written proofs and partial Lean verification	full geometric and computational classification is not established

Requirement	Status	Remaining gap
B-163 transfer	same-field fixed-family pullback checked	no smaller-field target or useful solver established
New affected family or lower-cost descent	unresolved	evaluated new route, source subgroup/log transfer, novelty and cost evidence

The next implementation step is an independently verified isogeny-path provider beyond Frobenius, starting from the complete small-field classes. Its inclusion must extend the declared universe explicitly. The open requirements are tracked in problems/transfer-audit-completeness.md, independently of PR completion.

Graphs checked: the existing GHS quartic-cubic transfer-scope graph and the candidate-catalogue curve/route definitions. They remain unchanged because no new curve identity, isogeny edge or weak family is certified here. These new figures describe proof scope and calculated bounds only.

Classification, obstruction and certificate trust

These are scoped consequences of established finite-group, abelian-variety and finite-field theory. They do not classify efficient algorithms or establish publication novelty. The word "transfer" needs the following distinctions.

1. Complete abstract prime-subgroup classification

Let S=<P> have prime order r, and let H be any finite abelian group. Evaluation at P gives a bijection Hom(S,H) -> H[r]. A homomorphism is injective precisely when its image of P is nonzero. Therefore an injective homomorphism exists if and only if r divides |H|.

Proof. A homomorphism must send P to R with rR=0, and is determined by that R. Conversely [j]P -> [j]R is well defined exactly when rR=0. A nonzero map from a prime-order group has zero kernel. Lagrange gives necessity of divisibility; Cauchy's theorem gives an element of order r for sufficiency. These arguments include r equal to the characteristic: H is the group of rational points, not a statement about geometric group schemes.

This classifies abstract homomorphisms, not cheaply evaluable maps. Evaluating the formula above from arbitrary source point coordinates may itself require the discrete logarithm j. Consequently r dividing a Jacobian order is not an efficient ECDLP reduction certificate.

2. Algebraic transfer classification up to isogeny

Let K=F_(q^N), E/K be an elliptic curve of trace t, W=Res_(K/k)(E), and A/k an abelian variety, where k=F_q. Write

P_W(T) = T^(2N) - t T^N + q^N.

Then Hom_k(W,A) tensor Q is nonzero if and only if gcd(P_W,P_A) over Q has positive degree. This applies to every k-algebraic group homomorphism between these varieties, without restricting equations, cover degree, or GHS family. Replacing E by a K-isogenous curve preserves this criterion.

Proof of the obstruction direction. A homomorphism commutes with Frobenius. By Cayley–Hamilton both characteristic polynomials annihilate its two Frobenius actions. If the polynomials are coprime, a rational Bezout identity, cleared of denominators, yields c*f=0 for some nonzero integer c. The image of f is connected and contained in the finite kernel of [c], so f=0. Equivalently the Hom group of abelian varieties is torsion free.

For the converse, decompose both varieties up to k-isogeny into simple factors. Tate's finite-field isogeny theorem identifies common irreducible Frobenius factors with a common simple isogeny factor. Projection onto and inclusion from that factor give a nonzero rational homomorphism; clearing denominators gives an actual nonzero homomorphism. Neither argument constructs a map, proves nonzero action on S, or verifies that A is a Jacobian.

In particular, if P_W is irreducible, every nonzero algebraic homomorphism W -> J(C) requires genus(C)>=N. Irreducibility/simplicity must be proved for the supplied instance; the CLI does not assume it. The implemented spectral test computes a primitive pseudo-remainder gcd over Q, conditionally on the supplied target polynomial being its actual Frobenius polynomial. It verifies reciprocity but does not certify an arbitrary polynomial is a Weil polynomial.

This theorem does not cover every map of the finite groups W(k) -> A(k): abstract homomorphisms need not extend to homomorphisms of abelian varieties. It also does not cover base-field enlargement or an unspecified target.

3. Uniform obstruction for all genus-bounded targets

Let S be an actual subgroup of order r>1 in E(K), and C/k any smooth, projective, geometrically connected curve of genus g. An injective group homomorphism S -> J(C)(k) requires

r <= #J(C)(k) <= (sqrt(q)+1)^(2g).

Thus if r>floor((sqrt(q)+1)^(2G)), no such transfer exists for any g<=G. This is independent of the cover family, source curve equation, and route through its K-isogeny class, for routes that preserve the same order-r subgroup. A route that kills that subgroup does not preserve its ECDLP.

Proof. An injection supplies r distinct rational Jacobian points. Each of the 2g complex Frobenius eigenvalues has absolute value sqrt(q), so the product formula for #J bounds each factor |1-alpha| by 1+sqrt(q). Monotonicity in g gives the uniform result. The same proof applies to abelian varieties of dimension g, not only Jacobians. It places no solver-cost or rho condition.

Exact integer evaluation uses (q+1+2 sqrt(q))^g=A_g+B_g sqrt(q), with

(A_0,B_0)=(1,0) A_(g+1)=(q+1) A_g+2q B_g B_(g+1)=2 A_g+(q+1) B_g.

The required floor is A_g+floor(sqrt(q B_g^2)). There are no floating-point approximations, including q=2. The CLI evaluates every subfield divisor.

For B-163 the certified prime order is 5846006549323611672814742442876390689256843201587. The exact bounds are

genus 63: 169721301146768189648370410887560301301009728819 7 genus 64: 98920823525104037575501729751467021228379369963 53.

Therefore every target over F2 of genus at most 63 is excluded. Genus 64 passes this necessary size test, which establishes no curve, map or attack. Same-field genus-three covers over F_(2^163) are unaffected by this F2 bound.

4. Finite classification and coverage trust theorem

Every ordinary elliptic curve over a binary finite field is isomorphic to $y^2+xy=x^3+a*x^2+b$, b!=0. Its j-invariant is 1/b. For a fixed b, transformations y -> y+s*x identify a modulo s^2+s. The image of s -> s^2+s is exactly the trace-zero subspace: its kernel has two elements, its image has codimension one and lies in that subspace. Thus a=0 and any fixed trace-one a give exactly the two twists for each b. An isomorphism preserves j, and the standard normal-form change with a1=1,a3=a4=0 reduces to the stated a equivalence, so these models exhaust ordinary Fq-isomorphism classes without duplicates. Two such elliptic curves are Fq-isogenous exactly when their orders/traces agree, by the elliptic case of Tate's theorem.

The CLI implements this complete enumeration only for degree<=8. Orders are counted by the Artin–Schreier trace criterion, with separate exhaustive x,y point-count controls through degree four. It does not manufacture paths between all equal-trace models. Frobenius coefficient orbits have evaluated paths, but do not exhaust a general isogeny class.

Trust theorem. For a declared finite universe U, enumeration V and rejection predicate R, suppose (a) U is contained in V, (b) the rejection test is sound for the precise transfer property claimed, and (c) every element of V is rejected. Then no element of U has that property. Proof: any purported element of U is in V by (a), rejected by (c), and excluded by (b).

The key obligations are (a) and (b), not the mere existence of a completed search log. The certificate verifier recomputes the canonical enumeration and its results. It cannot extend U to unimplemented transfer families or replace an unknown result with rejection. Hashes identify the implementation; they do not prove its correctness or authenticate the certificate's author.

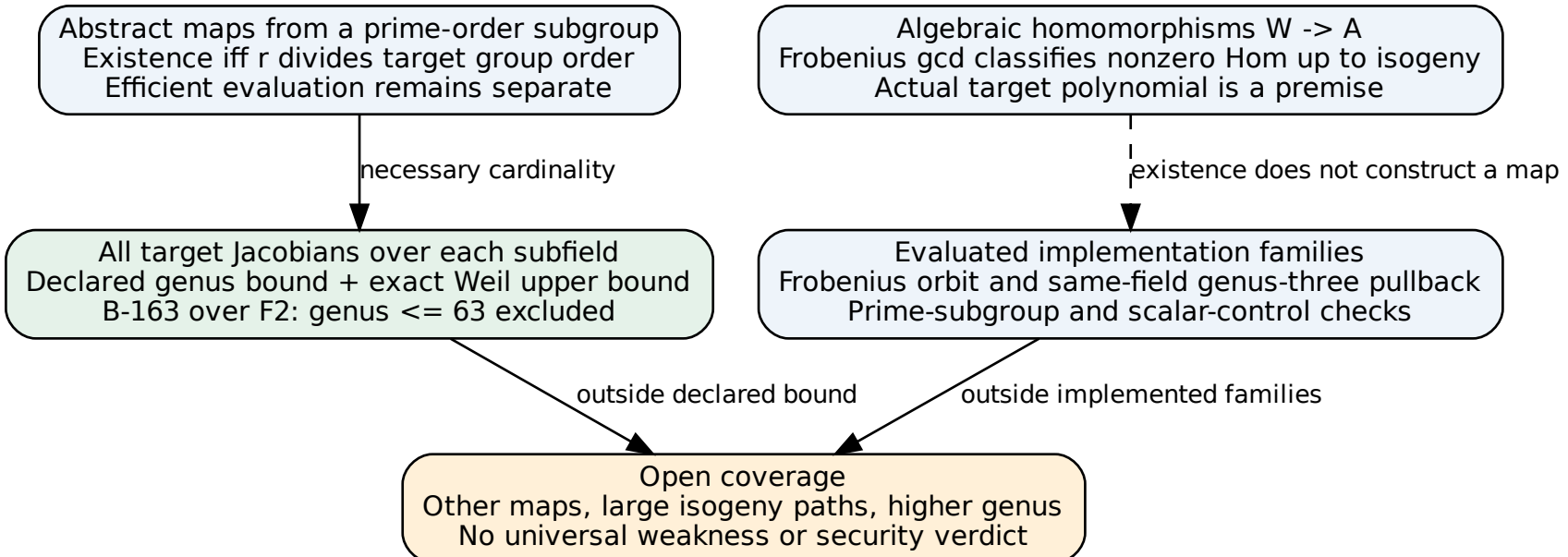
Proof map and trusted boundaries

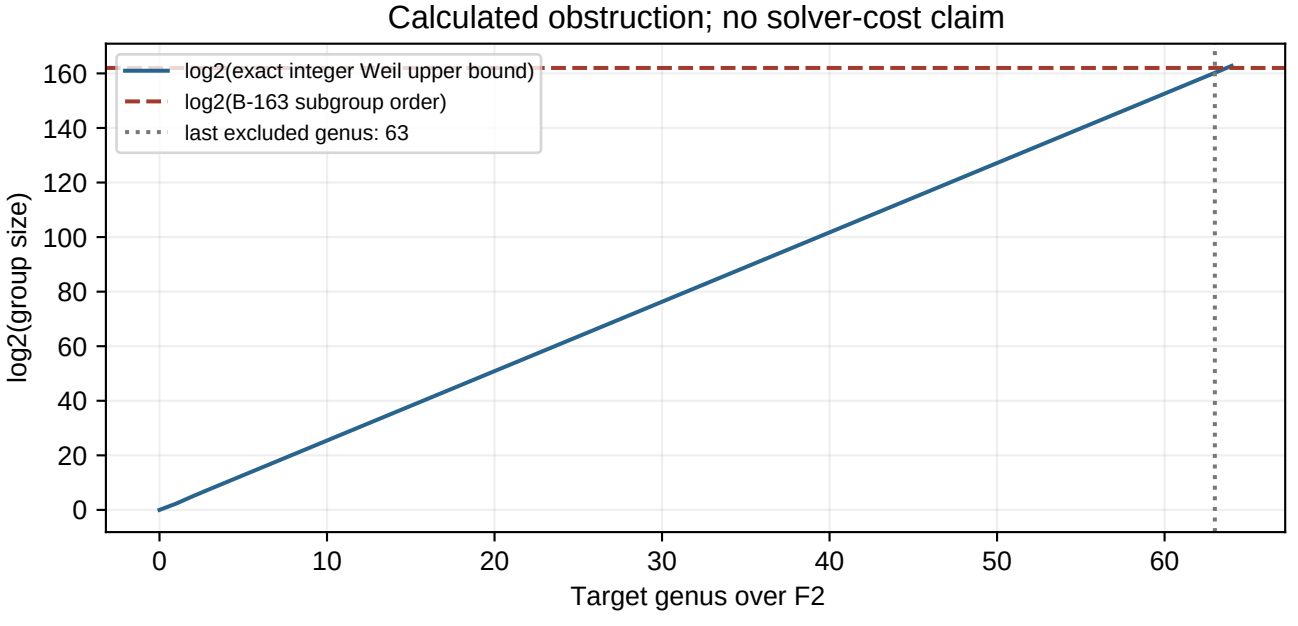
Claim	Lean coverage	Remaining bridge
Abstract Hom(S,H) classification	written proof only	finite abelian groups, Cauchy and efficient evaluation distinction
Frobenius algebraic classification	written proof only; exact gcd tested in Rust	abelian varieties, Frobenius, Tate and actual target polynomial
Uniform size obstruction	`size_obstruction`, `divisibility_obstruction`	injection/cardinality and geometric Weil bound
Exact B-163 integer threshold	`weilPair`, `sqrt_upper/lower`, `pair63/64`, `b163_genus63/64_bound`, `b163_obstruction`	identification with the geometric bound above; prime/order checked separately in Rust
Finite coverage implication	`closed_coverage`	enumerator completeness, rejection soundness and Rust/Lean refinement
Class-invariant transport	`invariant_transport`	trace-isogeny theorem and preservation of the chosen subgroup

Lean 4.34.1 checks eleven named theorems and one generated equation theorem transitively. Allowed foundations are propext, Classical.choice and Quot.sound; no custom axiom, sorry, native_decide, or kernel bypass is used. The complete mathematical/geometric classification is **not fully formalized**. This table retains the full statements and names the exact gaps rather than presenting the arithmetic lemmas as complete geometric proofs.

Sources and prior art

- Tate, *Endomorphisms of abelian varieties over finite fields*, Inventiones Mathematicae 2 (1966), 134–144, doi:10.1007/BF01404549; [primary text](https://pazuki.perso.math.cnrs.fr/index_fichiers/Tate66.pdf). - Menezes–Qu, *Analysis of the Weil descent attack of Gaudry, Hess and Smart*, CT-RSA 2001, doi:10.1007/3-540-45353-9_23, Corollary 9 and Table 2: M(163)=163, M(233)=30. This is a classical GHS statement, not a generalized obstruction to arbitrary covers. - [SEC 2 v2, section 3.2.3](https://www.secg.org/sec2-v2.pdf), B-163/sect163r2 field, coefficients, generator, prime order and cofactor. - [Magma Weil descent documentation](https://magma.maths.usyd.edu.au/magma/handbook/text/1561) describes GHS construction and divisor maps; parameter c changes the target. - The existing native same-field genus-three construction is documented in `suite/src/binary_ecc/cover_transfer.rs` and `.agents/skills/ecdlp-surfaces/references/higher-genus.md`. This audit reuses it. - Pocklington's n−1 criterion: the checked prime factors supply F|n−1, F²>n and the Fermat/gcd witnesses. For every prime l|n the witnesses force F|l−1, hence l>sqrt(n); n is therefore prime. This explains the verifier's exact certificate obligations independently of a probable-prime test.


Abstract maps from a prime-order subgroup
Existence iff r divides target group order
Efficient evaluation remains separate
Algebraic homomorphisms W -> A
Frobenius gcd classifies nonzero Hom up to isogeny
Actual target polynomial is a premise
necessary cardinality
existence does not construct a map
All target Jacobians over each subfield
Declared genus bound + exact Weil upper bound
B-163 over F2: genus <= 63 excluded
Evaluated implementation families
Frobenius orbit and same-field genus-three pullback
Prime-subgroup and scalar-control checks
outside declared bound
outside implemented families
Open coverage
Other maps, large isogeny paths, higher genus
No universal weakness or security verdict


Calculated obstruction; no solver-cost claim
log2(exact integer Weil upper bound)
log2(B-163 subgroup order)
last excluded genus: 63
log2(group size)
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50
60
Target genus over F2